Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++
Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu

Department of Computer Technology

Vision Statement of the Department.

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission Statement of the Department.

To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering ,innovation, research, and industry–institute collaboration & Holistic, ethical and sustainable development for lifelong success.

Session 2026-2027

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

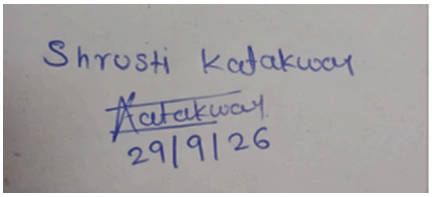
Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge,Problem analysis, Design/development of solutions,Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World,Ethics, Individual and Collaborative Team work,Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.



Name and Signature of Student and Date
(Signature and Date in Handwritten)

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Session	**2026-27 (ODD)**	**Course Name**	**HPCA**
Semester	**5th**	**Course Code**	**23ADS1524**
Roll No	**119**	**Name of Student**	**Shrusti Katakwar**

Practical Number	**04**
Course Outcome	1. Understand various internet technologies. 2. Design the web pages using HTML and CSS. 3. Implement the XML technology to store the data. 4. Develop the interactive web pages using JavaScript.
Aim	**Simulate pipelined computation using threads or message queues.**
Problem Definition	**Simulate pipelined computation using threads or message queues.**
Theory (100 words)	Pipelining is a technique in which a computation is divided into multiple stages, with each stage performing a specific operation. Each stage passes its result to the next stage using a queue or message-passing mechanism. In a multithreaded implementation, every stage can run in a separate thread, allowing different data items to be processed simultaneously. For example, Stage 1 may read data, Stage 2 may process it, and Stage 3 may generate the final output. This overlapping of operations improves processing speed, throughput, and CPU utilization. Message queues also provide safe communication between threads and help avoid data conflicts.

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Procedure and Execution (100 Words)	Step for Implementation: 1. **Divide the computation** into multiple sequential stages. 2. **Create a separate thread** for each pipeline stage. 3. **Create message queues** between consecutive stages. 4. **Send input data** to the first stage. 5. The first thread **processes the data** and sends the result to the next queue. 6. The next thread **receives and processes** the result. 7. Continue this process until the **final stage** produces the output. 8. Process multiple inputs simultaneously to demonstrate **pipelining**. 9. Use **synchronization** to avoid race conditions and data conflicts. 10. Send a **termination signal** after all inputs are processed. 11. Wait for all threads to complete. 12. **Measure execution time and throughput** and compare with sequential execution.
	Code: 

```
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <time.h>
#include <limits.h>

#define DATA_SIZE 20000
#define QUEUE_CAPACITY 100
#define POISON INT_MIN

typedef struct {
    int buffer[QUEUE_CAPACITY];
    int head;
    int tail;
    int count;
    pthread_mutex_t lock;
    pthread_cond_t not_full;
    pthread_cond_t not_empty;
} Queue;

void queue_init(Queue *q) {
    q->head = 0;
    q->tail = 0;
    q->count = 0;
    pthread_mutex_init(&q->lock, NULL);
    pthread_cond_init(&q->not_full, NULL);
    pthread_cond_init(&q->not_empty, NULL);
}

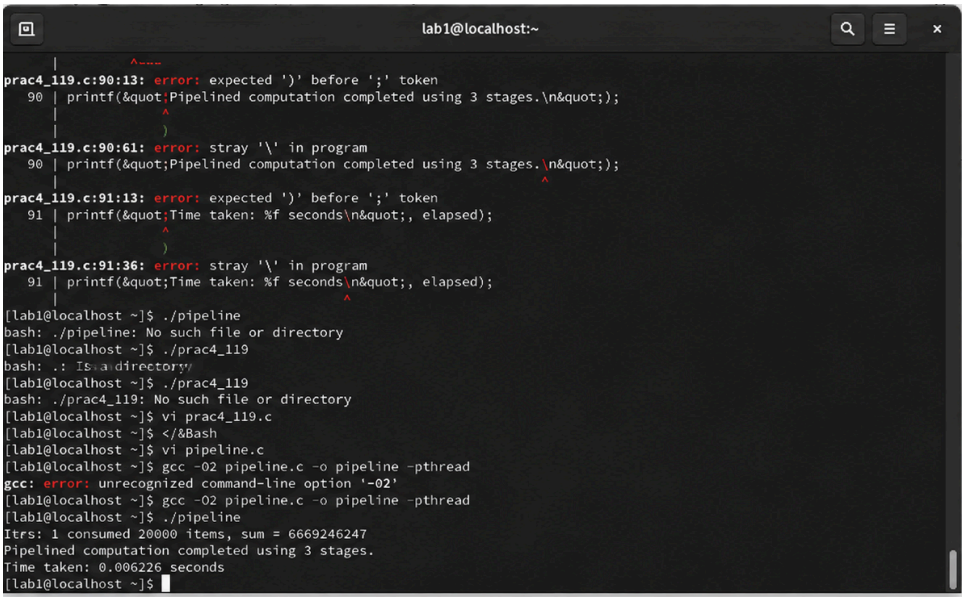
void queue_push(Queue *q, int value) {
    pthread_mutex_lock(&q->lock);
    while (q->count == QUEUE_CAPACITY) {
        pthread_cond_wait(&q->not_full, &q->lock);
    }
    q->buffer[q->tail] = value;
```

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	Output: 
Output Analysis	The pipelined computation successfully processes data through multiple stages using separate threads and message queues. Each stage performs its assigned operation and passes the result to the next stage. For example: Input → Stage 1 → Stage 2 → Stage 3 → Output 5 → 10 → 20 → 400 When multiple inputs are given, different stages work simultaneously. This reduces waiting time and improves throughput compared with sequential processing. The output confirms that data is correctly transferred between stages and processed in the proper order. Thus, the experiment demonstrates the effectiveness of parallel execution and inter-thread communication using pipelining.

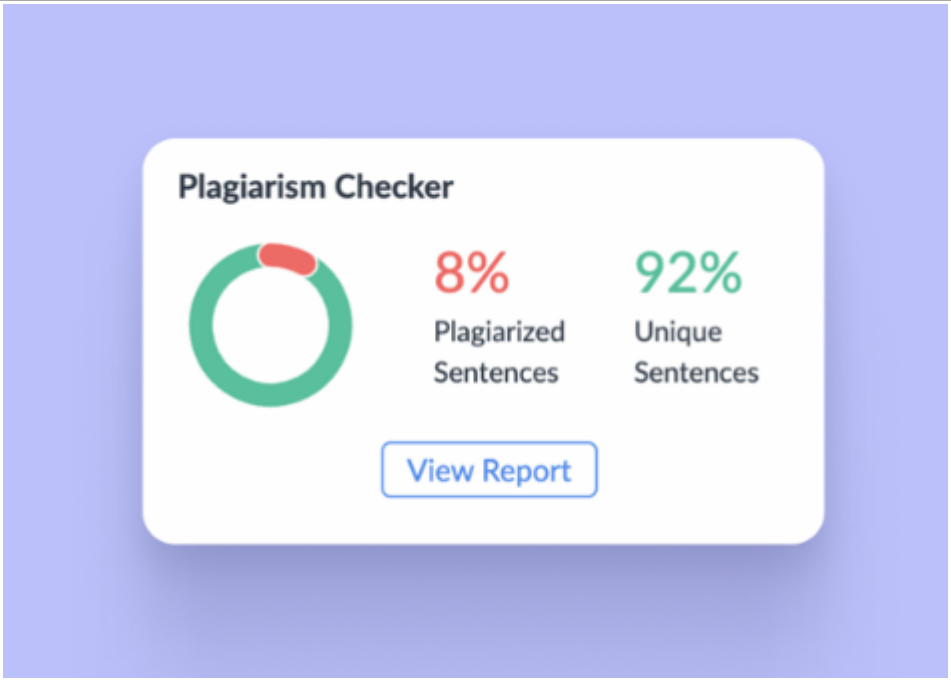
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Link of student Github profile where lab assignment has been uploaded	**https://github.com/Shrusti-Katakwar/HCPA**
Conclusion	The pipelined computation was successfully implemented using multiple threads and message queues. The computation was divided into different stages, allowing each stage to process data concurrently. Message queues provided effective communication between the stages. The experiment demonstrated that pipelining improves parallelism, throughput, and CPU utilization compared with sequential processing. Thus, pipelining is an efficient technique for processing multiple data items through a sequence of operations.
Plag Report (Similarity index < 12%)	
Date	**29/09/2026**